EXP.NO: 2	**Write JavaScript Programme to validate the following fields of the HTML Registration page.**
DATE: 06/02/2025	

AIM:

To write JavaScript code to validate user input in a registration form. The validation will ensure that the data entered by the user meets specific criteria for each field. This will help in maintaining data integrity and prevent invalid submissions.

ALGORITHM:

1. **Input:**
   - The user fills in the following fields: **First Name**, **Password**, **E-mail**, **Mobile Number**, **Last Name**, and **Address**.
2. **Validation Steps:**
   1. **Validate First Name:**
      - Check if the first name contains **only alphabets**.
      - Ensure the length of the **First Name** is **at least 6 characters**.
      - If any of these conditions fail, display an error message indicating the required input.
   2. **Validate Password:**
      - Check if the password is **at least 6 characters long**.
      - If it is less than 6 characters, display an error message.
   3. **Validate E-mail:**
      - Use a **regular expression** to check if the email follows the standard pattern: name@domain.com.
      - Ensure that the email contains **no invalid characters** and matches the required pattern.
      - If it doesn't match, display an error message indicating an invalid email format.

4. **Validate Mobile Number:**
    - Check if the mobile number contains **exactly 10 digits**.
    - Ensure there are **no alphabets or special characters**.
    - If the number is invalid, display an error message indicating the correct format.
5. **Validate Last Name:**
    - Ensure the **Last Name** field is **not empty**.
    - If it is empty, display an error message.
6. **Validate Address:**
    - Ensure the **Address** field is **not empty**.
    - If it is empty, display an error message.

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>University Result System | All-in-One</title>

  <style>
    :root {
      --sidebar-color: #2c3e50;
      --accent: #3498db;
      --success: #27ae60;
      --error-color: #e74c3c;
    }
```

```
body {
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    margin: 0;
    display: flex;
    height: 100vh;
    background-color: #eef2f3;
}

/* Side Frame Styling */
nav {
    width: 240px;
    background-color: var(--sidebar-color);
    color: white;
    padding-top: 20px;
    box-shadow: 2px 0 10px rgba(0,0,0,0.1);
}

nav h2 { text-align: center; font-size: 1.1rem; border-bottom: 1px solid #444; padding-bottom:
10px; }

.nav-link {
    display: block;
    padding: 15px 25px;
    color: #bdc3c7;
    text-decoration: none;
    cursor: pointer;
    transition: 0.3s;
}
```

```
.nav-link:hover {
    background-color: #34495e;
    color: white;
    padding-left: 35px;
}

/* Main Frame Styling */
main {
    flex: 1;
    padding: 30px;
    overflow-y: auto;
}

.container {
    background: white;
    max-width: 800px;
    margin: 0 auto;
    padding: 30px;
    border-radius: 12px;
    box-shadow: 0 5px 20px rgba(0,0,0,0.05);
}

/* Form & Input Styling */
.search-area, .form-group { display: flex; flex-direction: column; gap: 15px; margin-bottom: 25px; }

input[type="text"], input[type="password"], textarea {
```

```
    padding: 12px;
    border: 2px solid #ddd;
    border-radius: 6px;
    font-size: 14px;
}

button {
    padding: 12px 24px;
    background: var(--accent);
    color: white;
    border: none;
    border-radius: 6px;
    cursor: pointer;
    font-weight: bold;
    transition: 0.2s;
}

button:hover { opacity: 0.9; }

table { width: 100%; border-collapse: collapse; margin-top: 15px; }
th, td { padding: 12px; text-align: left; border-bottom: 1px solid #eee; }
th { background: #f8f9fa; color: #555; }

.error { color: var(--error-color); margin-top: 10px; font-weight: bold; font-size: 0.9rem; }

.map-box {
    border: 2px solid #eee;
    border-radius: 8px;
```

```
        overflow: hidden;
        margin-top: 20px;
      }
    </style>
  </head>
  <body>

    <nav>
      <h2>RESULT-PORTAL</h2>
      <a class="nav-link" onclick="toggleView('search')">Search Results</a>
      <a class="nav-link" onclick="toggleView('register')">Student Registration</a>
      <a class="nav-link" onclick="toggleView('campus')">Campus Map</a>
      <a class="nav-link" onclick="toggleView('support')">Support Hotspots</a>
      <hr style="border: 0.5px solid #444; margin: 10px 20px;">
      <a href="https://www.rajalakshmi.org/" target="_blank" class="nav-link">Official Website ↗</a>
    </nav>

    <main>
      <div class="container">

        <section id="search-view">
          <h1 style="color: var(--sidebar-color); border-bottom: 3px solid var(--accent); padding-bottom: 10px;">
            Examination 2025-2026
          </h1>

          <div class="search-area" style="flex-direction: row;">
            <input type="text" id="rollNo" placeholder="Enter Roll Number (147, 148, 149)">
```

```
            <button onclick="findResult()">Find Result</button>
        </div>

        <div id="errorMsg" class="error"></div>

        <div id="resultCard" style="display:none;">
            <div style="background: #f8f9fa; padding: 15px; border-radius: 8px; display: flex; justify-content: space-between;">
                <span><strong>Name:</strong> <span id="sName"></span></span>
                <span><strong>GPA:</strong> <span id="sGPA" style="color: var(--success); font-weight: bold;"></span></span>
            </div>
            <table>
                <thead>
                    <tr><th>Subject</th><th>Score</th><th>Grade</th></tr>
                </thead>
                <tbody id="resultTableBody"></tbody>
            </table>
        </div>
    </section>

    <section id="register-view" style="display:none;">
        <h1 style="color: var(--sidebar-color); border-bottom: 3px solid var(--success); padding-bottom: 10px;">Student Registration</h1>
        <div class="form-group">
            <input type="text" id="regFirstName" placeholder="First Name (Alphabets only, min 6 chars)">
            <input type="text" id="regLastName" placeholder="Last Name (Cannot be empty)">
            <input type="text" id="regEmail" placeholder="Email (name@domain.com)">
```

```
            <input type="password" id="regPass" placeholder="Password (min 6 chars)">
            <input type="text" id="regMobile" placeholder="Mobile Number (10 digits)">
            <textarea id="regAddress" placeholder="Address (Cannot be empty)" rows="3"></textarea>

            <button onclick="validateRegistration()" style="background: var(--success);">Submit Registration</button>
            <div id="regError" class="error"></div>
          </div>
        </section>

        <section id="campus-view" style="display:none;">
          <h1 style="color: var(--sidebar-color);">Campus Location</h1>
          <div class="map-box">
            <iframe src="https://www.google.com/maps/embed?pb=!1m18!1m12!1m3!1d3887.285375252872!2d79.95929!3d13.0142!2m3!1f0!2f0!3f0!3m2!1i1024!2i768!4f13.1!3m3!1m2!1s0x0%3A0x0!2zMTPCsDAwJzUxLjEiTiA3OcKwNTcnMzMuNCJF!5e0!3m2!1sen!2sin!4v1640000000000" width="100%" height="450" style="border:0;" allowfullscreen="" loading="lazy"></iframe>
          </div>
        </section>

        <section id="support-view" style="display:none;">
          <h1 style="color: var(--sidebar-color);">Support Hotspots</h1>
          <div class="map-box">
            <img src="rec_logo.png" usemap="#helpmap" style="width: 100%;" alt="Support Zones">
            <map name="helpmap">
              <area shape="rect" coords="0,0,400,200" alt="Academic" href="javascript:void(0)" onclick="alert('Navigating to Academic Support...')">
```

```
                <area shape="rect" coords="401,0,800,200" alt="Tech" href="javascript:void(0)" onclick="alert('Opening Tech Support Chat...')">
            </map>
        </div>
    </section>

  </div>
</main>

<script>
  // MOCK DATABASE
  const database = {
    "149": { name: "Naresh kumar", gpa: "3.85", scores: [{ sub: "OTA", mark: 88, grade: "A" }, { sub: "Web Technology", mark: 96, grade: "B+" }, { sub: "Software Construction", mark: 92, grade: "A+" }] },
    "147": { name: "Sri sidtharthan", gpa: "3.60", scores: [{ sub: "OTAI", mark: 75, grade: "B+" },{ sub: "Web Technology", mark: 93, grade: "B+" }, { sub: "Software Construction", mark: 89, grade: "A" }] },
    "148": { name: "Prasham Jaganathan", gpa: "4.00", scores: [{ sub: "OTAI", mark: 98, grade: "A+" }, { sub: "Web Technology", mark: 95, grade: "B+" }, { sub: "Software Construction", mark: 100, grade: "A+" }] }
  };

  // VIEW TOGGLE LOGIC
  function toggleView(view) {
    const views = ['search-view', 'register-view', 'campus-view', 'support-view'];
    views.forEach(v => {
      document.getElementById(v).style.display = (v === view + '-view') ? 'block' : 'none';
    });
  }
```

```
// FIND RESULT LOGIC
function findResult() {
    const input = document.getElementById('rollNo').value.trim();
    const card = document.getElementById('resultCard');
    const error = document.getElementById('errorMsg');
    const tableBody = document.getElementById('resultTableBody');

    error.innerText = "";
    card.style.display = "none";
    tableBody.innerHTML = "";

    if (database[input]) {
        const student = database[input];
        document.getElementById('sName').innerText = student.name;
        document.getElementById('sGPA').innerText = student.gpa;

        student.scores.forEach(item => {
            tableBody.innerHTML += `<tr><td>${item.sub}</td><td>${item.mark}</td><td>${item.grade}</td></tr>`;
        });
        card.style.display = "block";
    } else {
        error.innerText = "Invalid Roll Number. Please use 147, 148, or 149.";
    }
}

// REGISTRATION VALIDATION LOGIC
```

```
function validateRegistration() {
    const fName = document.getElementById('regFirstName').value.trim();
    const lName = document.getElementById('regLastName').value.trim();
    const email = document.getElementById('regEmail').value.trim();
    const pass = document.getElementById('regPass').value;
    const mobile = document.getElementById('regMobile').value.trim();
    const address = document.getElementById('regAddress').value.trim();
    const errorDiv = document.getElementById('regError');

    errorDiv.innerText = "";

    // a) First Name: Alphabets only and length >= 6
    const alphaRegex = /^[A-Za-z]+$/;
    if (fName.length < 6 || !alphaRegex.test(fName)) {
        errorDiv.innerText = "Error: First Name must contain only alphabets and be at least 6 characters long.";
        return;
    }

    // b) Password: Length >= 6
    if (pass.length < 6) {
        errorDiv.innerText = "Error: Password must be at least 6 characters long.";
        return;
    }

    // c) E-mail: Standard pattern name@domain.com
    const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (!emailRegex.test(email)) {
```

```
            errorDiv.innerText = "Error: Invalid Email format (use name@domain.com).";
            return;
        }

        // d) Mobile Number: Exactly 10 digits
        const mobileRegex = /^\d{10}$/;
        if (!mobileRegex.test(mobile)) {
            errorDiv.innerText = "Error: Mobile Number must contain exactly 10 digits.";
            return;
        }

        // e) Last Name and Address: Not Empty
        if (lName === "" || address === "") {
            errorDiv.innerText = "Error: Last Name and Address cannot be empty.";
            return;
        }

        alert("Registration Successful!");
        // Optional: Reset form after success
        document.querySelectorAll('#register-view input, #register-view textarea').forEach(el => el.value = "");
    }
  </script>
</body>
</html>
```

OUTPUT:

RESULT:

HTML page with Form is created and is validated using Javascript